

JEE MAIN REVISION MASTER: ENVIRONMENTAL CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 5 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Atmospheric Pollution, Smog & Green Chemistry	5 PYQs	Environmental Science

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Atmospheric Pollution, Smog & Green Chemistry	$\text{Acid Rain: } \text{pH} < 5.6, \quad \text{Photochemical Smog} = \text{O}_3 + \text{PAN} + \text{Acrolein} + \text{NO}_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Atmospheric Pollution, Smog & Green Chemistry (Moderate Yield — 5 PYQs)

Core Principle: Classical smog: reducing smog, occurs in cool humid climate, contains smoke, fog and SO₂.

Photochemical smog: oxidizing smog, occurs in warm sunny climate, contains unsaturated hydrocarbons and nitrogen oxides, forming ozone, acrolein, and PAN (peroxyacetyl nitrate). Acid rain: pH < 5.6 caused by SO₂ and NO_x.

Water quality: Clean drinking water has BOD < 5 ppm; highly polluted water has BOD > 17 ppm.

■ **Caution / JEE Pitfall:** Photochemical smog is OXIDIZING in nature, while classical smog is REDUCING in nature!

■ **Speed Trick:** Chlorofluorocarbons (CFCs) deplete ozone in the stratosphere via chlorine free radical chain catalysis: one Cl radical can destroy up to 100,000 ozone molecules.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The first chlorinated organic insecticide prepared is..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "Identify the incorrect statement from the following...." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Which of the following is not a green house gas?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#2)

Identify the incorrect statement from the following.

- (A) Oxides of nitrogen in the atmosphere can cause depletion of the ozone layer.
- (B) Ozone absorbs the intense ultraviolet radiation of Sun.
- (C) Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes.
- (D) Ozone absorbs infrared radiation.

Correct Answer: None

Solution Analysis: Option D is incorrect. Ozone primarily absorbs ultraviolet (UV) radiation, particularly in the wavelength range of 240 to 320 nanometers, which helps to protect living organisms from harmful UV rays emitted by the sun. It does not absorb infrared radiation significantly; instead, gases such as carbon dioxide (CO_2), water vapor (H_2O), and methane (CH_4) are primarily responsible for the absorption of infrared radiation, contributing to the greenhouse effect.